

Proof That Displacement Interpolation Gives Wasserstein Geodesics

Assume $\mu, \nu \in \mathcal{P}_2(\mathbb{R}^d)$ and that the optimal transport for the quadratic cost is induced by a Brenier map

$$T : \mathbb{R}^d \rightarrow \mathbb{R}^d, \quad T_{\#}\mu = \nu.$$

For the quadratic cost, define

$$T_t(x) = (1-t)x + tT(x), \quad t \in [0, 1],$$

and

$$\mu_t = (T_t)_{\#}\mu.$$

This curve moves each particle along the straight line from x to $T(x)$.

Claim

The curve $(\mu_t)_{t \in [0,1]}$ is a constant-speed geodesic:

$$W_2(\mu_s, \mu_t) = |t - s| W_2(\mu, \nu), \quad s, t \in [0, 1].$$

Proof

Fix $0 \leq s < t \leq 1$. Consider the coupling

$$\gamma_{s,t} = (T_s, T_t)_{\#}\mu \in \Pi(\mu_s, \mu_t).$$

This coupling pairs the position at time s and the position at time t of the same original particle x . For each such particle,

$$|T_t(x) - T_s(x)| = |(t-s)(T(x) - x)| = |t-s| |T(x) - x|.$$

Using $\gamma_{s,t}$ as an admissible transport plan gives

$$W_2^2(\mu_s, \mu_t) \leq \int_{\mathbb{R}^d \times \mathbb{R}^d} |a - b|^2 d\gamma_{s,t}(a, b).$$

We now rewrite this integral in terms of the original variable x . Define

$$F_{s,t}(x) = (T_s(x), T_t(x)).$$

By definition,

$$\gamma_{s,t} = (F_{s,t})_{\#}\mu.$$

The [defining property](#) of a pushforward measure says that, for every integrable function $\Phi : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$,

$$\int_{\mathbb{R}^d \times \mathbb{R}^d} \Phi(a, b) d\gamma_{s,t}(a, b) = \int_{\mathbb{R}^d} \Phi(F_{s,t}(x)) d\mu(x).$$

Here we choose

$$\Phi(a, b) = |a - b|^2.$$

Therefore

$$\int_{\mathbb{R}^d \times \mathbb{R}^d} |a - b|^2 d\gamma_{s,t}(a, b) = \int_{\mathbb{R}^d} |T_s(x) - T_t(x)|^2 d\mu(x).$$

This is why the integral over $\mathbb{R}^d \times \mathbb{R}^d$ becomes an integral over $\mathbb{R}^d$: the coupling $\gamma_{s,t}$ was itself obtained by pushing the original measure μ forward through the map $x \mapsto (T_s(x), T_t(x))$.

Now compute the integrand:

$$T_t(x) - T_s(x) = ((1 - t)x + tT(x)) - ((1 - s)x + sT(x)).$$

The terms involving x and $T(x)$ collect as

$$T_t(x) - T_s(x) = (s - t)x + (t - s)T(x) = (t - s)(T(x) - x).$$

Hence

$$|T_t(x) - T_s(x)|^2 = |t - s|^2 |T(x) - x|^2.$$

Substituting this into the previous integral gives

$$\int_{\mathbb{R}^d \times \mathbb{R}^d} |a - b|^2 d\gamma_{s,t}(a, b) = |t - s|^2 \int_{\mathbb{R}^d} |T(x) - x|^2 d\mu(x).$$

Thus

$$W_2^2(\mu_s, \mu_t) \leq |t - s|^2 \int_{\mathbb{R}^d} |T(x) - x|^2 d\mu(x).$$

Since T is optimal from μ to ν ,

$$\int_{\mathbb{R}^d} |T(x) - x|^2 d\mu(x) = W_2^2(\mu, \nu).$$

Thus

$$W_2(\mu_s, \mu_t) \leq |t - s| W_2(\mu, \nu).$$

It remains to prove the reverse inequality. Apply the same upper bound on the intervals $[0, s]$ and $[t, 1]$:

$$W_2(\mu, \mu_s) \leq s W_2(\mu, \nu), \quad W_2(\mu_t, \nu) \leq (1 - t) W_2(\mu, \nu).$$

By the triangle inequality,

$$W_2(\mu, \nu) \leq W_2(\mu, \mu_s) + W_2(\mu_s, \mu_t) + W_2(\mu_t, \nu).$$

Therefore

$$W_2(\mu, \nu) \leq s W_2(\mu, \nu) + W_2(\mu_s, \mu_t) + (1 - t) W_2(\mu, \nu).$$

Rearranging,

$$W_2(\mu_s, \mu_t) \geq |t - s| W_2(\mu, \nu).$$

Combining the upper and lower bounds gives

$$W_2(\mu_s, \mu_t) = |t - s| W_2(\mu, \nu).$$

This is exactly the constant-speed geodesic condition.

Interpretation

Euclidean geodesics move points along straight lines. Wasserstein geodesics move mass along optimal straight-line particle trajectories. This is why the curve $\mu_t = (T_t)_\# \mu$ is called displacement interpolation.